The documentation and process conversion measures necessary to comply with this revision shall be completed by 20 March 2004.

INCH-POUND

MIL-PRF-19500/614D
20 December 2004
SUPERSEDING
MIL-PRF-19500/614C
8 September 2003

* PERFORMANCE SPECIFICATION SHEET

# SEMICONDUCTOR DEVICE, FIELD EFFECT RADIATION HARDENED TRANSISTOR, N-CHANNEL, SILICON, TYPES 2N7380 AND 2N7381, JANTXV M, D, R, F, G, AND H, JANS M, D, R, F, G, AND H

This specification is approved for use by all Departments and Agencies of the Department of Defense.

* The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

## 1. SCOPE

* 1.1 Scope. This specification covers the performance requirements for an N-channel, radiation hardened, enhancement mode, MOSFET, power transistor intended for use in high density power switching applications. Two levels of product assurance are provided for each device type as specified in MIL-PRF-19500, with avalanche energy ratings ($E_{AS}$) and maximum avalanche current ($I_{AS}$).

1.2 Physical dimensions. See figure 1 (TO-257AA).

* 1.3 Maximum ratings. Unless otherwise specified, $T_C$ = +25°C.

| Type | Min $V_{(BR)DSS}$ $V_{GS}$ = 0 V $I_D$ = 1.0 mA dc | $P_T$ (1) $T_C$ = +25°C | $P_T$ $T_A$ = +25°C (free air) | $R_{\theta JC}$ | $I_{D1}$ (2) (3) $T_C$ = +25°C | $I_{D2}$ (2) (3) $T_C$ = +100°C | $T_J$ and $T_{STG}$ |
|---|---|---|---|---|---|---|---|
| | V dc | W | W | °C/W | A dc | A dc | °C |
| 2N7380 | 100 | 75 | 2 | 1.67 | 14.4 | 9.1 | -55 to +150 |
| 2N7381 | 200 | 75 | 2 | 1.67 | 9.4 | 6.0 | -55 to +150 |

| Type | $I_S$ | $I_{DM}$ (4) | Max $r_{DS(on)}$ (1) $V_{GS}$ = 12 V dc $I_D = I_{D2}$ | | $V_{GS}$ | $E_{AS}$ max | $I_{AS}$ |
|---|---|---|---|---|---|---|---|
| | | | $T_J$ = +25°C | $T_J$ = +150°C | | | |
| | A dc | A(pk) | Ω | Ω | V dc | mJ | A dc |
| 2N7380 | 14.4 | 57 | 0.18 | 0.33 | ±20 | 150 | 14.4 |
| 2N7381 | 9.4 | 37 | 0.40 | 0.84 | ±20 | 150 | 9.4 |

(See notes next page)

* Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil/.



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* 1.3 Maximum ratings. continued

(1) Derate linearly by 0.6 W/°C for $T_C$ > +25°C;
(2) The following formula derives the maximum theoretical $I_D$ limit. $I_D$ is limited by package and device construction.

$$I_D = \sqrt{\frac{T_{JM} - T_C}{(R_{\theta JC}) \times (R_{DS}(on) \text{ at } T_{JM})}}$$

(3) See figure 2, maximum drain current graph.
(4) $I_{DM}$ = 4 X $I_{D1}$ as calculated in note (2).

1.4 Primary electrical characteristics. Unless otherwise specified, $T_C$ = +25°C.

| Type | Min $V_{(BR)DSS}$ $V_{GS}$ = 0 $I_D$ = 1.0 mA dc | $V_{GS(th)1}$ $V_{DS} \geq V_{GS}$ $I_D$ = 1.0 mA dc | | $I_{DSS}$ max $V_{GS}$ = 0 $V_{DS}$ = 80 percent of rated $V_{DS}$ | Max $r_{DS(on)1}$ (1) $V_{GS}$ = 12 V; $I_D$ = $I_{D2}$ $T_J$ = +25°C |
|---|---|---|---|---|---|
| | V dc | V dc | | μA dc | Ω |
| | | Min | Max | | |
| 2N7380 | 100 | 2.0 | 4.0 | 25 | 0.18 |
| 2N7381 | 200 | 2.0 | 4.0 | 25 | 0.40 |

(1) Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

* 2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

* 2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

* DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

* DEPARTMENT OF DEFENSE STANDARDS

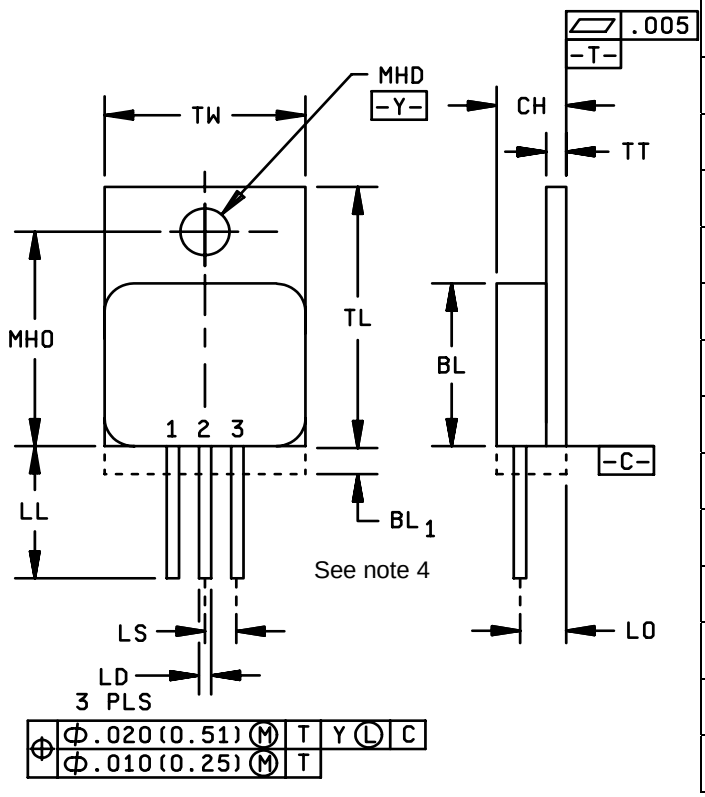
MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.



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| Ltr | Inches | | Millimeters | |
|---|---|---|---|---|
| | Min | Max | Min | Max |
| BL | .410 | .420 | 10.41 | 10.67 |
| $BL_1$ | | .033 | | 0.84 |
| CH | .190 | .200 | 4.83 | 5.08 |
| LD | .025 | .035 | 0.64 | 0.89 |
| LL | .600 | .650 | 15.24 | 16.51 |
| LO | .120 BSC | | 3.05 BSC | |
| LS | .100 BSC | | 2.54 BSC | |
| MHD | .140 | .150 | 3.56 | 3.81 |
| MHO | .527 | .537 | 13.39 | 13.64 |
| TL | .645 | .665 | 16.38 | 16.89 |
| TT | .035 | .045 | 0.89 | 1.14 |
| TW | .410 | .420 | 10.41 | 10.67 |
| Term 1 | Drain | | | |
| Term 2 | Source | | | |
| Term 3 | Gate | | | |

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. All terminals are isolated from case.
4. This area is for the lead feed-thru eyelets (configuration is optional, but will not extend beyond this zone).
* 5. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

* FIGURE 1. Dimensions and configuration (TO-257AA).



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3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

* 3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

$I_{AS}$ ..... Rated avalanche current, nonrepetitive.
nC .....nano Coulomb

* 3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1 (TO-257AA). Methods used for electrical isolation of the terminal feedthroughs shall employ materials that contain a minimum of 90 percent $AL_2O_3$ (ceramic). Examples of such construction techniques are metallized ceramic eyelets or ceramic walled packages.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Internal construction. Multiple chip construction shall not be permitted.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.6 Electrostatic discharge protection. The devices covered by this specification require electrostatic protection.

3.6.1 Handling. MOS devices must be handled with certain precautions to avoid damage due to the accumulation of static charge. The following handling procedures shall be followed:

a. Devices shall be handled on benches with conductive handling devices.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent, if practical.

g. Care shall be exercised, during test and troubleshooting, to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source. $R \leq 100\ k\Omega$, whenever bias voltage is to be applied drain to source.

3.7 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.8 Electrical test requirements. The electrical test requirements shall be table I as specified herein.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.



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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

* c. Conformance inspection (see 4.4 and tables I and II).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

* 4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.



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4.3 Screening (JANS and JANTXV levels only). Screening shall be in accordance with table IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen (see table IV of MIL-PRF-19500) (1) (2) | Measurement | |
|---|---|---|
| | JANS level | JANTXV level |
| (3) | Gate stress test (see 4.3.1) | Gate stress test (see 4.3.1) |
| (3) | Method 3470 of MIL-STD-750. (see 4.3.2) | Method 3470 of MIL-STD-750. (see 4.3.2) |
| (3) 3c | Method 3161 of MIL-STD-750 (see 4.3.3) | Method 3161 of MIL-STD-750 (see 4.3.3) |
| 7 | Optional. | Optional. |
| 9 | $I_{GSSF1}$, $I_{GSSR1}$, $I_{DSS1}$, subgroup 2 of table I herein; | Not applicable |
| 10 | Method 1042 of MIL-STD-750, test condition B | Method 1042 of MIL-STD-750, test condition B |
| 11 | $I_{GSSF1}$, $I_{GSSR1}$, $I_{DSS1}$, $r_{DS(on)1}$, $V_{GS(th)1}$<br>Subgroup 2 of table I herein.<br>$\Delta I_{GSSF1}$ = ±20 nA dc or ± 100 percent of initial value, whichever is greater.<br>$\Delta I_{GSSR1}$ = ±20 nA dc or ± 100 percent of initial value, whichever is greater.<br>$\Delta I_{DSS1}$ = ±25 μA dc or ± 100 percent of initial value, whichever is greater. | $I_{GSSF1}$, $I_{GSSR1}$, $I_{DSS1}$, $r_{DS(on)1}$, $V_{GS(th)1}$<br>Subgroup 2 of table I herein. |
| 12 | Method 1042 of MIL-STD-750, test condition A<br>t = 240 hours | Method 1042 of MIL-STD-750, test condition A |
| 13 | Subgroup 2 and 3 of table I herein.<br>$\Delta I_{GSSF1}$ = ±20 nA dc or ± 100 percent of initial value, whichever is greater.<br>$\Delta I_{GSSR1}$ = ±20 nA dc or ± 100 percent of initial value, whichever is greater.<br>$\Delta I_{DSS1}$ = ±25 μA dc or ± 100 percent of initial value, whichever is greater.<br><br>$\Delta r_{DS(on)1}$ = ±20 percent of initial value.<br>$\Delta V_{GS(th)1}$ = ±20 percent of initial value. | Subgroup 2 of table I herein.<br>$\Delta I_{GSSF1}$ = ±20 nA dc or ± 100 percent of initial value, whichever is greater.<br>$\Delta I_{GSSR1}$ = ±20 nA dc or ± 100 percent of initial value, whichever is greater.<br>$\Delta I_{DSS1}$ = ±25 μA dc or ± 100 percent of initial value, whichever is greater.<br><br>$\Delta r_{DS(on)1}$ = ±20 percent of initial value.<br>$\Delta V_{GS(th)1}$ = ±20 percent of initial value. |
| 14 | Required. | Required. |

(1) At the end of the test program, $I_{GSSF1}$, $I_{GSSR1}$, and $I_{DSS1}$ are measured.
(2) An out-of-family program to characterize $I_{GSSF1}$, $I_{GSSR1}$, $I_{DSS1}$, and $V_{GS(th)1}$ shall be invoked.
(3) Shall be performed anytime before screen 9.



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4.3.1 Gate stress test. Apply $V_{GS}$ = ±24 V minimum for t = 250 μs minimum.

4.3.2 Single pulse avalanche energy ($E_{AS}$).

a. Peak current ($I_{AS}$)..... $I_{D1}$.

b. Peak gate voltage ($V_{GS}$)..... 12 V.

c. Gate to source resistor ($R_{GS}$)..... $25 \leq R_{GS} \leq 200\ \Omega$.

d. Initial case temperature ..... +25°C +10°C, -5°C.

e. Inductance ..... $\left[\frac{2E_{AS}}{(I_{D1})^2}\right]\left[\frac{V_{BR} - V_{DD}}{V_{BR}}\right]$ mH minimum.

f. Number of pulses to be applied ..... 1 pulse minimum.

g. Supply voltage ..... $V_{DD}$ = 50 V, or 25 V for 100 V devices.

4.3.3 Thermal impedance . The thermal impedance measurements shall be performed in accordance with method 3161 of MIL-STD-750. The maximum limit (not to exceed figure 3, thermal impedance curves and the table I, subgroup 2 limits) for thermal impedance in screening (table IV of MIL-PRF-19500) shall be derived by each vendor by means of statistical process control. When the process has exhibited control and capability, the capability data shall be used to establish the fixed screening limit. In addition to screening, once a fixed limit has been established, monitor all future sealing lots using a random five piece sample from each lot to be plotted on the applicable X bar R chart. If a lot exhibits an out of control condition, the entire lot shall be removed from the line and held for engineering evaluation and disposition. This procedure may be used in lieu of an in line process monitor.

a. Measuring current ($I_M$) ..... 10 mA.

b. Drain heating current ($I_H$) ..... 2 A minimum.

c. Heating time ($t_H$) ..... 50 ms.

d. Drain-source heating voltage ($V_H$) ..... 15 V minimum.

e. Measurement time delay ($t_{MD}$) ..... 30 μs to 60 μs maximum.

f. Sample window time ($t_{SW}$)..... 10 μs maximum.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (JANS) and table VIb (JANTXV) of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.



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4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| B3 | 1051 | Condition G. |
| B4 | 1042 | The heating cycle shall be 1 minute minimum, 2,000 cycles. No heat sink nor forced air cooling on the device shall be permitted. |
| B5 | 1042 | Condition A; $V_{DS}$ = 100 percent of rated; $T_A$ = +175°C, t = 120 hours or $T_A$ = +150°C, t = 240 hours; read and record $V_{BR(DSS)}$ (pre and post) at $I_D$ = 1 mA; read and record $I_{DSS}$ (pre and post), in accordance with table I, subgroup 2. |
| B5 | 1042 | Condition B; $V_{GS}$ = 100 percent of rated $T_A$ = +175°C, t = 24 or $T_A$ = +150°C, t = 48 hours. |
| B6 | 3161 | See 4.5.2. |

4.4.2.2 Group B inspection, table VIb (JANTXV) of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| B2 | 1051 | Condition G. |
| B3 | 1042 | The heating cycle shall be 1 minute minimum. |

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

| Subgroup | Method | Condition |
|---|---|---|
| C2 | 2036 | Test condition A, weight = 10 lbs, t = 10 seconds. |
| C5 | 3161 | See 4.5.2. |
| * C6 | 1042 | The heating cycle shall be 1 minute minimum, 6,000 cycles. No heat sink nor forced air cooling on the device shall be permitted. |

4.4.4 Group D Inspection. Group D inspection shall be conducted in accordance with appendix E, table VIII of MIL-PRF-19500 and table II herein.

4.4.5 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table IX of MIL-PRF-19500 and as specified herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.



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4.5 Methods of inspection. Methods of inspection shall be as specified in appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurements shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance measurements shall be performed in accordance with method 3161 of MIL-STD-750. The maximum limit of $R_{\theta JC(max)}$ shall be 1.67°C/W. The following parameter measurements shall apply:

a. Measuring current ($I_M$)..... 10 mA.

b. Drain heating current ($I_H$) ..... 2 A minimum.

c. Heating time ($t_H$)..... Steady-state (see method 3161 of MIL-STD-750 for definition).

d. Drain-source heating voltage ($V_H$)..... 15 V minimum.

e. Measurement time delay ($t_{MD}$)..... 30 μs to 60μs maximum.

f. Sample window time ($t_{SW}$)..... 10 μs maximum.



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* TABLE I. Group A inspection.

| | Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|
| | | Method | Condition | | Min | Max | |
| | Subgroup 1 | | | | | | |
| | Visual and mechanical inspection | 2071 | | | | | |
| | Subgroup 2 | | | | | | |
| | Thermal impedance 2/ | 3161 | See 4.3.3 | $Z_{\theta JC}$ | | 1.30 | °C/W |
| | Breakdown voltage drain to source | 3407 | $V_{GS}$ = 0, $I_D$ = 1 mA dc, bias condition C | $V_{(BR)DSS}$ | | | |
| | 2N7380 | | | | 100 | | V dc |
| | 2N7381 | | | | 200 | | V dc |
| | Gate to source voltage (threshold) | 3403 | $V_{DS} \geq V_{GS}$, $I_D$ = 1.0 mA | $V_{GS(th)1}$ | 2.0 | 4.0 | V dc |
| * | Gate current | 3411 | $V_{GS}$ = +20 V dc, $V_{DS}$ = 0, bias condition C | $I_{GSSF1}$ | | +100 | nA dc |
| * | Gate current | 3411 | $V_{GS}$ = -20 V dc, $V_{DS}$ = 0, bias condition C | $I_{GSSR1}$ | | -100 | nA dc |
| | Drain current | 3413 | $V_{GS}$ = 0, $V_{DS}$ = 80 percent of rated $V_{DS}$, bias condition C | $I_{DSS1}$ | | 25 | µA dc |
| | Static drain to source on-state resistance | 3421 | $V_{GS}$ = 12 V dc, condition A, pulsed (see 4.5.1), $I_D$ = rated $I_{D2}$ (see 1.3) | $r_{DS(on)1}$ | | | |
| | 2N7380 | | | | | 0.18 | Ω |
| | 2N7381 | | | | | 0.40 | Ω |
| | Static drain to source on-state resistance | 3421 | $V_{GS}$ = 12 V dc, condition A, pulsed (see 4.5.1), $I_D$ = rated $I_{D1}$ (see 1.3) | $r_{DS(on)2}$ | | | |
| | 2N7380 | | | | | 0.20 | Ω |
| | 2N7381 | | | | | 0.49 | Ω |
| | Forward voltage (source drain diode) | 4011 | $V_{GS}$ = 0, $I_D$ = rated $I_{D1}$ pulsed (see 4.5.1) | $V_{SD}$ | | | |
| | 2N7380 | | | | | 1.8 | V dc |
| | 2N7381 | | | | | 1.4 | V dc |

See footnotes at end of table.



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* TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Condition | | Min | Max | |
| Subgroup 3 | | | | | | |
| High temperature operation: | | $T_A$ = +125°C | | | | |
| Gate current | 3411 | Bias condition C, $V_{GS}$ = ±20 V dc, $V_{DS}$ = 0 | $I_{GSS2}$ | | ±200 | nA dc |
| Drain current | 3413 | Bias condition C, $V_{GS}$ = 0, $V_{DS}$ = 80 percent of rated $V_{DS}$ | $I_{DSS3}$ | | 0.25 | mA dc |
| Static drain to source on-state | 3421 | $V_{GS}$ = 12 V dc, pulsed (see 4.5.1), $I_D$ = rated $I_{D2}$ | $r_{DS(on)3}$ | | | |
| 2N7380 | | | | | 0.35 | Ω |
| 2N7381 | | | | | 0.75 | Ω |
| Gate to source voltage (threshold) | 3403 | $V_{DS} \geq V_{GS}$, $I_D$ = 1.0 mA dc | $V_{GS(th)2}$ | 1.0 | | V dc |
| Low temperature operation: | | $T_A$ = -55°C | | | | |
| Gate to source voltage (threshold) | 3403 | $V_{DS} \geq V_{GS}$, $I_D$ = 1.0 mA dc | $V_{GS(th)3}$ | | 5.0 | V dc |
| Subgroup 4 | | | | | | |
| Switching time test | 3472 | $I_D$ = rated $I_{D1}$, $V_{GS}$ = 12 V dc, gate drive impedance = 7.5 Ω, $V_{DD}$ = 50 percent of $V_{BR(DSS)}$ | | | | |
| Turn-on delay time | | | $t_{d(on)}$ | | 25 | ns |
| Rise time | | | $t_r$ | | | |
| 2N7380 | | | | | 60 | ns |
| 2N7381 | | | | | 50 | ns |
| Turn-off delay time | | | $t_{d(off)}$ | | | |
| 2N7380 | | | | | 40 | ns |
| 2N7381 | | | | | 70 | ns |
| Fall time | | | $t_f$ | | | |
| 2N7380 | | | | | 30 | ns |
| 2N7381 | | | | | 60 | ns |

See footnotes at end of table.



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* TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Condition | | Min | Max | |
| Subgroup 4 - Continued. | | | | | | |
| Forward transconductance | 3475 | $I_D = I_{D2}$, $V_{DD}$ = 15 V dc see 4.5.1 | $g_{fs}$ | 2.5 | | s |
| Subgroup 5 | | | | | | |
| Safe operating area test (high voltage) | 3474 | See figure 4; $t_p$ = 10 ms, $V_{DS}$ = 80 percent of rated $V_{BR(DSS)}$, $V_{DS}$ = 200 V maximum | | | | |
| Electrical measurements | | See table I, subgroup 2 | | | | |
| Subgroup 6 | | | | | | |
| Not applicable | | | | | | |
| Subgroup 7 | | | | | | |
| Gate charge | 3471 | Condition B | | | | |
| On-state gate charge | | | $Q_{g(on)}$ | | | |
| 2N7380 | | | | | 40 | nC |
| 2N7381 | | | | | 50 | nC |
| Gate to source charge | | | $Q_{gs}$ | | 10 | nC |
| Gate to drain charge | | | $Q_{gd}$ | | | |
| 2N7380 | | | | | 20 | nC |
| 2N7381 | | | | | 25 | nC |
| Reverse recovery time | 3473 | $d_i/d_t \leq 100$ A/µs, $V_{DD} \leq 50$ V, $I_D = I_{D1}$ | $t_{rr}$ | | | |
| 2N7380 | | | | | 275 | ns |
| 2N7381 | | | | | 460 | ns |

1/ For sampling plan, see MIL-PRF-19500.
2/ This test is required for the following end-point measurement only (not intended for 4.3, screen 9, 11, or 13): JANS, table VIa of MIL-PRF-19500, group B, subgroups 3 and 4; JANTXV, table VIb of MIL-PRF-19500, group B, subgroups 2 and 3; and table VII of MIL-PRF-19500, group C, subgroup 6, and table IX of MIL-PRF-19500, group E, subgroup 1.



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TABLE II. Group D inspection.

| Inspection 1/ 2/ 3/ | MIL-STD-750 | | Symbol | Pre-irradiation limits | | | | Post-irradiation limits | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Method | Conditions | | M, D, and R | | F, G, and H 4/ | | M, D, and R | | F, G, and H 4/ | | |
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| Subgroup 2 | | $T_C$ = +25°C | | | | | | | | | | |
| Steady-state total dose irradiation ($V_{GS}$ bias) 5/ | 1019 | $V_{GS}$ = 12 V, $V_{DS}$ = 0 | | | | | | | | | | |
| Steady-state total dose irradiation ($V_{DS}$ bias) 5/ | 1019 | $V_{GS}$ = 0, $V_{DS}$ = 80 percent of rated $V_{DS}$ (pre-irradiation) | | | | | | | | | | |
| End-point electrical: | | | | | | | | | | | | |
| Breakdown voltage, drain to source | 3407 | $V_{GS}$ = 0, $I_D$ = 1 mA, bias condition C | $V_{(BR)DSS}$ | | | | | | | | | |
| 2N7380 | | | | 100 | | 100 | | 100 | | 100 | | V dc |
| 2N7381 | | | | 200 | | 200 | | 200 | | 200 | | V dc |
| Gate to source voltage 4/ (threshold) | 3403 | $V_{DS} \geq V_{GS}$, $I_D$ = 1 mA | $V_{GS(th)}$ | 2.0 | 4.0 | 2.0 | 4.0 | 2.0 | 4.0 | 1.25 | 4.5 | V dc |
| Gate current | 3411 | $V_{GS}$ = 20 V, $V_{DS}$ = 0 V, bias condition C | $I_{GSSF1}$ | | 100 | | 100 | | 100 | | 100 | nA dc |
| Gate current | 3411 | $V_{GS}$ = 20 V, $V_{DS}$ = 0, bias condition C | $I_{GSSR1}$ | | -100 | | -100 | | -100 | | -100 | nA dc |

See footnotes at end of table.



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TABLE II. Group D inspection - Continued.

| Inspection 1/ 2/ 3/ | MIL-STD-750 | | Symbol | Pre-irradiation limits | | | | Post-irradiation limits | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Method | Conditions | | M, D, and R | | F, G, and H 4/ | | M, D, and R | | F, G, and H 4/ | | |
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| Subgroup 2 - Continued | | $T_C$ = +25°C | | | | | | | | | | |
| Drain current | 3413 | $V_{GS}$ = 0, bias condition C, $V_{DS}$ = 80 percent of rated $V_{DS}$ (pre-irradiation) | $I_{DSS}$ | | | | | | | | | |
| 2N7380 | | | | | 25 | | 25 | | 25 | | 50 | µA dc |
| 2N7381 | | | | | 25 | | 25 | | 25 | | 50 | µA dc |
| Static drain to source on-state voltage | 3405 | $V_{GS}$ = 12 V, condition A pulsed, see 4.5.1. $I_D = I_{D2}$ | $V_{DS(ON)}$ | | | | | | | | | |
| 2N7380 | | | | | 1.638 | | 1.638 | | 1.638 | | 2.184 | V dc |
| 2N7381 | | | | | 2.4 | | 2.4 | | 2.4 | | 3.18 | V dc |
| Forward voltage source drain diode | 4011 | $V_{GS}$ = 0, $I_D = I_{D1}$, bias condition C | $V_{SD}$ | | | | | | | | | |
| 2N7380 | | | | | 1.8 | | 1.8 | | 1.8 | | 1.8 | V |
| 2N7381 | | | | | 1.4 | | 1.4 | | 1.4 | | 1.4 | V |

1/ For sampling plan, see MIL-PRF-19500.
2/ Separate samples shall be pulled for each bias.
3/ Group D qualification may be performed anytime prior to lot formation. Wafers qualified to these group D QCI requirements may be used for any other specification sheet utilizing the same die design.
4/ The F designation represents devices which pass end-points at both 100K and 300K rads (Si). The G designation represents devices which pass 100K, 300K and 600K rad (Si) end-points.
5/ H must meet end points for 300K and 1,000K rad (Si).



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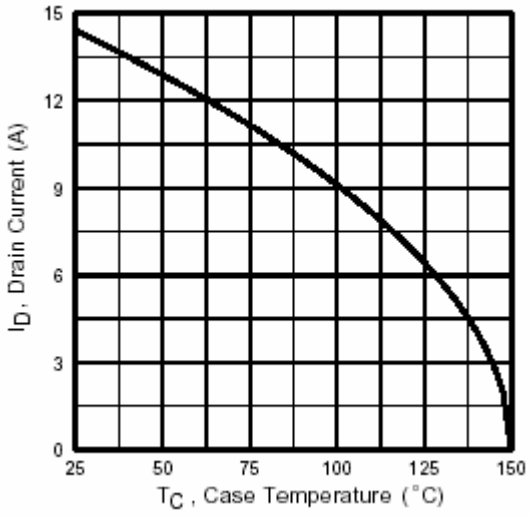
* TABLE III. Group E inspection (all quality levels) - for qualification or re-qualification only.

| | Inspection | MIL-STD-750 | | Qualification and large lot quality conformance inspection |
|---|---|---|---|---|
| | | Method | Conditions | |
| | Subgroup 1 | | | 12 devices, c = 0 |
| | Temperature cycling | 1051 | Test condition G, 500 cycles | |
| * | Hermetic seal<br>Fine leak<br>Gross leak | 1071 | <br>Test conditions G or H<br>Test conditions C or D | |
| | Electrical measurements | | See table I, subgroup 2 | |
| | Subgroup 2 1/ | | | 12 devices, c = 0 |
| | Steady-state reverse bias | 1042 | Condition A, 1,000 hours | |
| | Electrical measurements | | See table I, subgroup 2 | |
| | Steady-state gate bias | 1042 | Condition B, 1,000 hours | |
| * | Subgroup 4 | | | Sample size N/A |
| | Thermal impedance curves | | Each supplier shall submit their qual-lot average and design maximum thermal impedance curves to the qualifying activity. In addition, the optimal test conditions and thermal impedance limit shall be provided to the qualifying activity in the qualification report. | |
| | Subgroup 5 | | | |
| | Not applicable | | | |
| | Subgroup 6 | | | 3 devices |
| * | ESD | 1020 | Not required for devices classified as ESD class 1. | |
| * | Subgroup 8 | | | 22 devices, c = 0 |
| | Commutating diode for safe operating area test procedure for measuring dv/dt during reverse recovery of power MOSFET transistors or insulated gate bipolar transistors | 3476 | | |

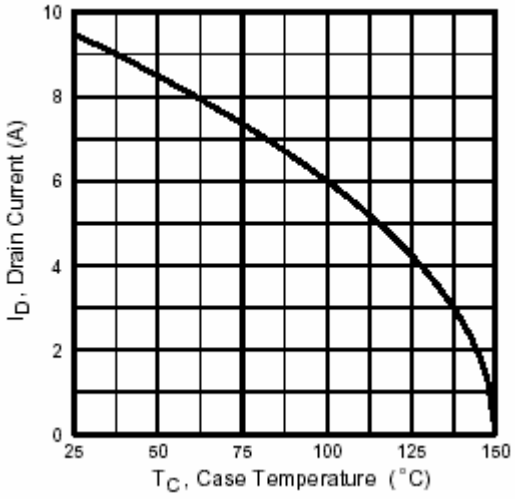
1/ A separate sample for each test may be pulled.



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2N7380




2N7381

FIGURE 2. Maximum drain current vs case temperature graphs.



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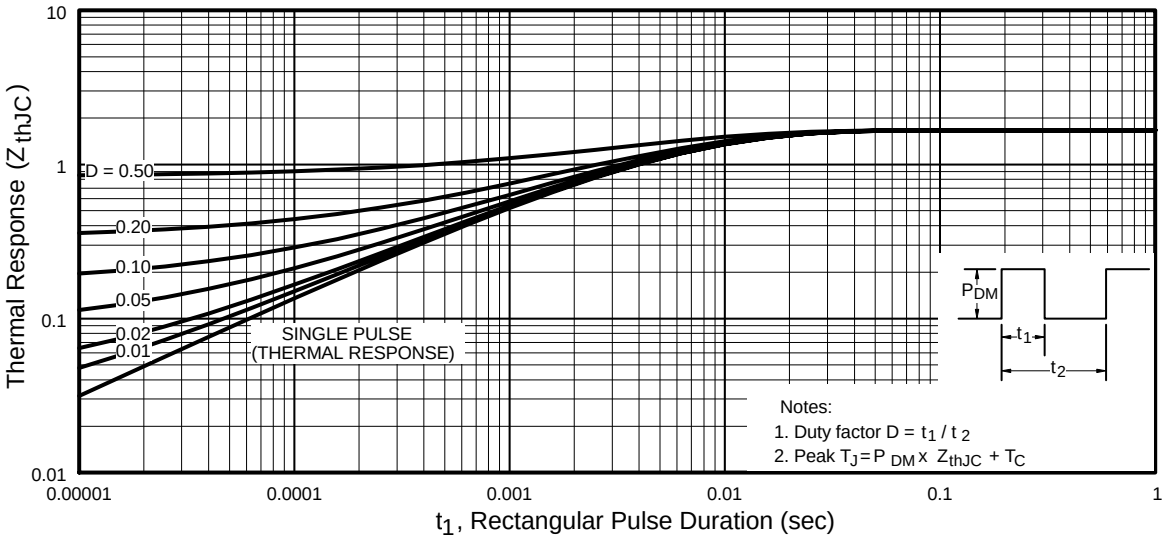



FIGURE 3. Thermal response curves.



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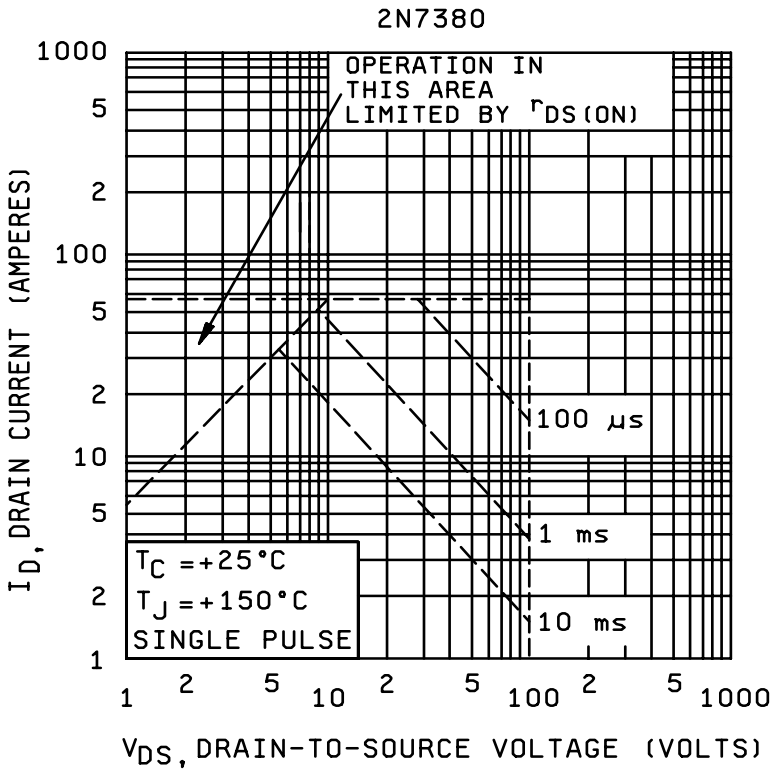



FIGURE 4. Safe operating area graphs.



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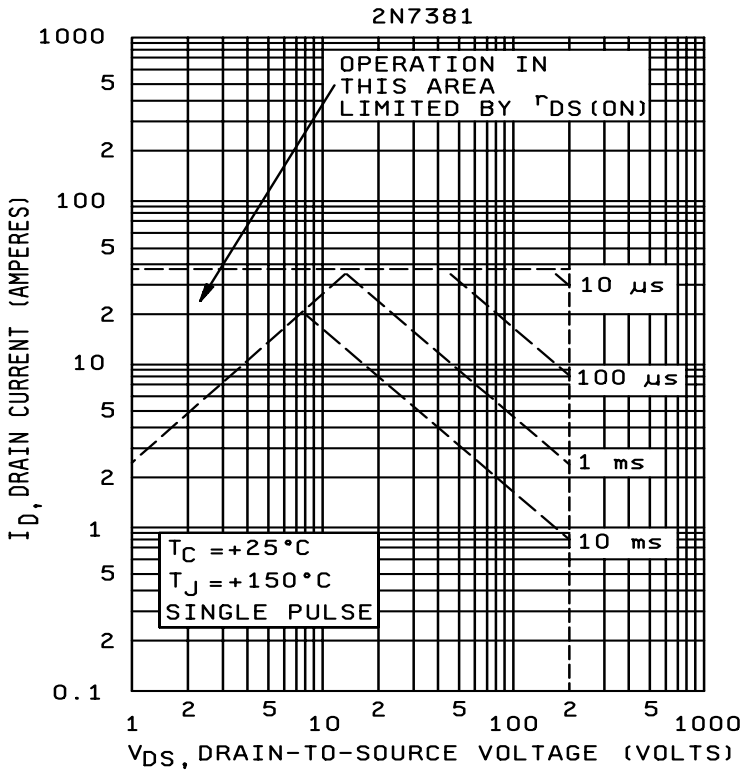



FIGURE 4. Safe operating area graphs - Continued.



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5. PACKAGING

* 5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

* 6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil.

6.4 Supersession data. This specification supersedes DESC drawing 89009, dated 19 December 1989.



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6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Navy - EC
Air Force - 11
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Review activities:
Army - SM
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Preparing activity:
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